PROJECT PROPOSAL: INTRODUCTION TO MACHINE LEARNING

1. Team Information

- **Team Name:** Three-Sigma
- **Members & Roles:**
  - **Bakhramov Abdulaziz - Modelling Lead**: Responsible for implementing 3+ models and cross-validation.
  - **Abdullayev Zhanibek - EDA Lead**: Responsible for visualisations, missing value analysis, and feature engineering.
  - **Aliyev Karim - Report & Presentation Lead**: Responsible for the 5-7 page final report and 5-minute presentation.

2. Dataset Selection

- **Name:** Student Performance
- **Source URL:** https://archive.ics.uci.edu/dataset/320
- **Dimensions:** 649 rows and 33 columns
- **Verification:** Dataset has been downloaded and successfully opened in Python.

3. Problem Definition

- **Problem Type:** Regression
- **Target Variable:** `G3` (Final Grade)

4. Hypothesis

- **Hypothesis:** We predict that `failures` (number of past class failures) will be the strongest negative predictor of the final grade, while `studytime` will show a significant positive correlation with academic success.

5. Planned Feature Approach

- **Feature 1: `failures`** — Previous academic setbacks often indicate underlying difficulties with the subject matter, strongly impacting the final result.
- **Feature 2: `absences`** — Higher number of missed classes directly correlates with missing key material, which we expect to lower the `G3` score.
- **Categorical Encoding:** We identified that features such as `Mjob` (mother's job) and `school` are categorical and will require **One-Hot Encoding** in Week 2.